Juhyun Jung

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INTERESTS

Bio-inspired Robotics, Robotics, Deep Learning

EDUCATION

M.S. / ***Mechanical Eng.***
Stanford University | Stanford, CA — Sep 2025 – Present
(4.06/4.0 GPA)

B.S. / ***Mechanical Eng. (Minor: Artificial Intelligence, College of Computer Science)***
Illinois Institute of Technology | Chicago, IL — Aug 2022 - Dec 2024
Dean's list (4.0/4.0 GPA)

B.S. / ***Mechanical Eng.***
Ajou University | Korea — Mar 2018 - Dec 2024
Dean's list (4.47/4.5 GPA) *Ranked 1st in class (1/170)*

RELEVANT COURSEWORK & SKILLS

Robotics: Robot Autonomy I(CS237A), Collaborative Robotics (ME326), Intro to Robotics (CS223), Optimal and Learning-Based Control (AA203)
CS: Deep learning (CS230), Decision Making Uncertainty (CS238), Deep Reinforcement Learning (CS224R)
Programming: Python, ROS2, Linux Ubuntu 22.04, MATLAB, Julia, C/C++

RESEARCH EXPERIENCE

Biomimetics and Dextrous Manipulation Lab(BDML) / Stanford University| Stanford, CA — Sep 2025 – Present
Advisor: Dr. Mark Cutkosky

UMI-underwater: Robust Manipulation ***(ongoing)*** — Apr 2026 – Present

- Developed and implemented robust control policies for an underwater manipulator to achieve stable grasping under hydrodynamic disturbances.

Whisker-shaped Underwater Fiber-Bragg-Grating (FBG) Tactile Sensor ***(under submission)*** — Sep 2025 – Apr 2026

- Developed a novel tactile sensing modality inspired by animal whiskers, integrating FBG optical strain sensors with compliant beams for underwater flow detection.
- Achieved >8× improvement in signal-to-noise ratio through experimental design and signal processing.
- Designed and built an embedded experimental platform with stepper-motor actuation and real-time data acquisition.

Computer Vision and Multimedia Laboratory (CVM lab) / Illinois Institute of Technology | Chicago, IL — June 2023 – Dec 2024
Advisor: Dr. Yan Yan

- Developed an algorithm using LSTM to classify mouse behavior based on pose dataset of mice.
- Created a behavior classification algorithm using Vision Transformer (ViT) applied to entire mouse video datasets, enhancing behavior prediction accuracy.
- Assisted PhD students in evaluating the performance of multimodal models, including Concatenation Fusion and Gated Information Fusion, for behavior classification tasks.

Multiscale Bio-inspired Technology Laboratory (MOST Lab) / Ajou University | South Korea — Jan 2022 – June 2022
Advisor: Dr. Je-sung Koh

- Designed and developed a comprehensive experimental setup for advanced DEA research.
- Conducted extensive experiments manipulating key parameters such as frequency, electrode area, elastomer composition, thickness, voltage, and pre-strain.
- Achieved a significant breakthrough, demonstrating a maximum DEA strain of 12.3%.

PROJECTS

Understanding the varied economic impact of Tax Increment Financing (TIF) Jan 2024 – May 2024
Advisor: Dr. Robert Ellis / Consulting Expert: Tom Tresser (CivicLab Co-Founder)

- Analyzed the impact of Tax Increment Financing (TIF) on median income trends in Chicago, revealing a significant increase for the White population in TIF districts, while other racial groups remained unaffected.
- Revealed that White-dominant, wealthy regions experienced a significantly higher median income increase compared to other areas.

Autonomous Mobility Competition - Vision Lane Following Project Sep 2021 – Dec 2021

- Achieved 2nd place in a competition by HL Mando Corporation by designing and implementing a vision-based lane following algorithm, utilizing advanced image processing and PID control to optimize real-time data for improved model car performance.

System Innovation competition Mar 2021 – May 2021
Advisor: Dr. Peom Park

- Awarded 3rd place in the Korean Society of Systems Engineering (KOSSE) competition for analyzing limitations of VR technology such as cybersickness and designing solutions like accelerometer integration and sound effects to enhance user experience, supported by comprehensive market surveys.

PROFESSIONAL PRESENTATIONS

Poster Presentation / Illinois Institute of Technology | Chicago, IL Nov 2024
"Autonomous control using automated image analysis"
Fall 2024 Amour R&D Expo, Chicago, IL

Poster Presentation / Illinois Institute of Technology | Chicago, IL July 2024
"Autonomous control using automated image analysis"
Summer 2024 Amour R&D Research Immersion Exposition, Chicago, IL

Poster Presentation / Illinois Institute of Technology | Chicago, IL April 2024
"Understanding the varied economic impact of Tax Increment Financing (TIF)"
Ed Kaplan Family Institute for Innovation and Tech Entrepreneurship Innovation Day, Chicago, IL

Poster Presentation / The Korean Society of Systems Engineering | South Korea June 2021
"Assessment of Cybersickness in VR Social Media Services"
The Korean Society of Systems Engineering 2021 Fall Conference, Jeju, South Korea

OTHER ACTIVITIES

- Sigma Phi Epsilon Fraternity, Illinois Beta – The Balanced Man Scholarship Chair
- Tau Beta Pi, The Engineering Honor Society
- Climbing Club

HONORS AND AWARDS

Dean's List	Illinois Institute of Technology	2022 - 2024
Dean's List	Ajou University	2018,2021,2022
Academic Excellence Scholarship	Ajou University	2018,2021,2022
Department Chair's Award for Academic Excellence	Ajou University	2019,2022
Mando Autonomous mobility VLF (Vision Lane Following) 2nd award	HL Mando Corporation	2021
System Innovation Competition 3rd award	The Korean Society of Systems Engineering	2021
Soldier of Valor Award	Republic of Korea Army	2020
Active Learner Self-Development Scholarship	Ajou University	2018-Spring, 2018-Fall